

Rubico Tech — Full Stack Developer

Candidate Technical Task

Estimated Time: 2-3 days

Expected Experience: 4+ years

Submission: GitHub/Gitlab or any other publicly available repository link shared with HR, ethan.ankur@rubicotech.in, and ed.arun@rubicotech.in.

AI-Powered Event & Marketplace Platform

Problem Statement

Build a **scalable platform** where:

- Organizations can create **events** and **products**.
 - Users can **browse, search (natural language)**, and **interact (like/save/register)**.
 - The system must support **AI-powered semantic search**. **Bonus: recommendations for events and products**.
-

Core Requirements

1. Backend (API + Logic)

Identify and design the entities and their schemas and relationships.

Required APIs:

- Auth for org users (simple JWT is enough)
- CRUD for Events & Products
- Feed API:
 - Returns mixed list (events + products)

- Search API:
 - Accepts **natural language query**
 - Returns relevant events/products
-

2. Frontend (UI)

Create a simple UI (React / Next.js preferred):

Pages:

- Login/Register
- Organization management
- Events/Products management for organization users.
- Feed page
- Search page

Features:

- Display mixed feed (events + products)
- Like/save toggle
- Search input (natural language)
- Show relevance-ranked results

Focus is NOT on pixel perfection, but:

- State management
 - API integration
 - UX clarity
-

3. Code Expectations

- Production quality
 - Best practices followed
 - Test cases included for critical parts
 - Proper git history
-

4. Deliverables

- GitHub repo
- README with:

- Setup instructions
 - Architecture decisions
 - Tradeoffs
- Attach output/link of any and all AI tools used to build the project.
- **Bonus:**
 - API documentation (swagger or any other)
 - CI/CD pipeline
 - Containerized code